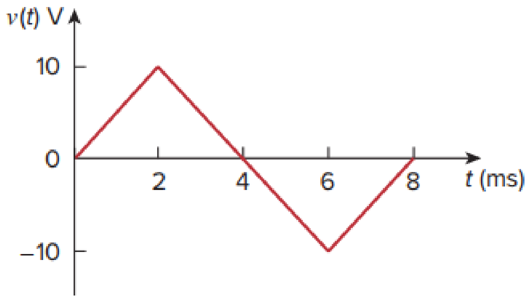
Chapter 6, Problem 5P

Problem

The voltage across a 4-μF capacitor is shown in Fig. 6.45. Find the current waveform.

Fig. 6.45:



Step-by-step solution

Step 1 of 7

Refer to Figure 6.45 in the textbook.

Consider the formula for straight line equation for two points (x_1, y_1) and (x_2, y_2).

$$(y - y_1) = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

Where,

$$y = v(t)$$
$$x = t$$

Step 2 of 7

For $0 \le t < 2\text{ ms}$, calculate the straight line equation of points $(0,0)$ and $(2\text{m},10)$.

$$(y - 0) = \frac{10 - 0}{2\times10^{-3} - 0}(x - 0)$$

$$y = \frac{10}{2\times10^{-3}}x$$

$$y = 5000x$$

$$v(t) = 5000t \qquad \text{for } 0 \le t < 2\text{ ms}$$

Step 3 of 7

For $2\text{ ms} \le t < 6\text{ ms}$, calculate the straight line equation of points $(2\text{m},10)$ and $(6\text{m},-10)$

$$(y-10) = \frac{-10-10}{6\times10^{-3}-2\times10^{-3}}\left(x-2\times10^{-3}\right)$$

$$(y-10) = \frac{-20}{4\times10^{-3}}\left(x-2\times10^{-3}\right)$$

$$y-10 = -5000x+10$$
$$y = -5000x+20$$

$$v(t) = -5000t+20 \qquad \text{for } 2\text{ ms} \le t < 6\text{ ms}$$

Step 4 of 7

For $6\text{ ms} \le t < 8\text{ ms}$, calculate the straight line equation of points $(6\text{m},-10)$ and $(8\text{m},0)$.

$$(y-(-10)) = \frac{0+10}{8\times10^{-3}-6\times10^{-3}}\left(x-6\times10^{-3}\right)$$

$$y+10 = 5000\left(x-6\times10^{-3}\right)$$
$$y+10 = 5000x-30$$
$$y = 5000x-40$$

$$v(t) = 5000t-40 \qquad \text{for } 6\text{ ms} \le t < 8\text{ ms}$$

Step 5 of 7

Therefore, the voltage values are,

$$v(t) = \begin{cases} 5000t & 0 < t < 2\text{ ms} \\ 20-5000t & 2\text{ ms} < t < 6\text{ ms} \\ -40+5000t & 6\text{ ms} < t < 8\text{ ms} \end{cases}$$

Write the expression for the current through the capacitor.

$$i(t) = C\frac{dv}{dt}$$

Calculate the current through the current through the capacitor for $0 < t < 2\text{ ms}$.

$$\begin{aligned} i(t) &= \left(4\times10^{-6}\right)\frac{d}{dt}(5000t) \\ &= \left(4\times10^{-6}\right)(5000) \\ &= 20\times10^{-3} \\ &= 20\text{ mA} \end{aligned}$$

Step 6 of 7

Calculate the current through the current through the capacitor for $2\text{ ms} < t < 6\text{ ms}$.

$$\begin{aligned} i(t) &= \left(4\times10^{-6}\right)\frac{d}{dt}(20-5000t) \\ &= \left(4\times10^{-6}\right)(0-5000) \\ &= -20\times10^{-3} \\ &= -20\text{ mA} \end{aligned}$$

Calculate the current through the current through the capacitor for $6\text{ ms} < t < 8\text{ ms}$.

$$\begin{aligned} i(t) &= \left(4\times10^{-6}\right)\frac{d}{dt}(-40+5000t) \\ &= \left(4\times10^{-6}\right)(0+5000) \\ &= 20\times10^{-3} \\ &= 20\text{ mA} \end{aligned}$$

Step 7 of 7

Draw the current through the capacitor waveform:

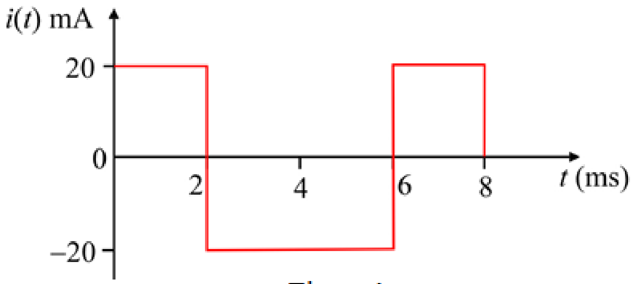


Figure 1